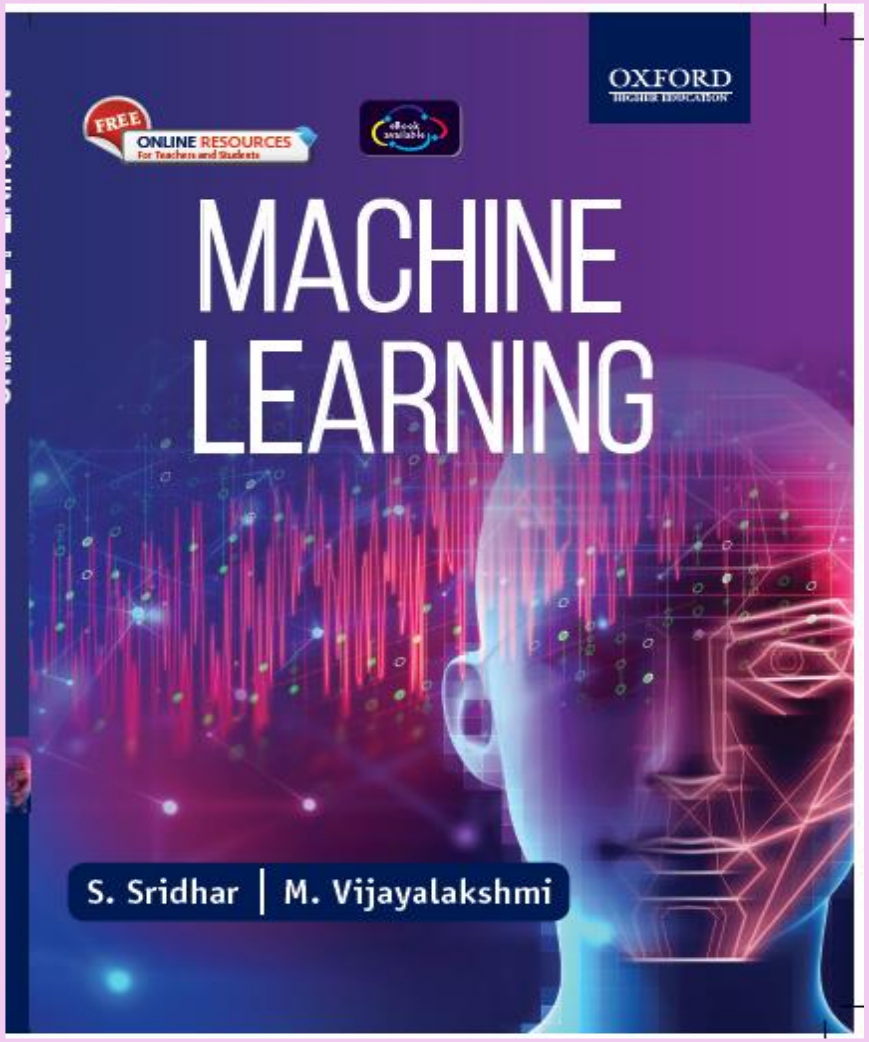


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Machine Learning

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Chapter 8

Bayesian Learning

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Bayesian Learning

Bayesian Learning is a learning method that describes and represents knowledge in an uncertain domain and provides a way to reason about this knowledge using probability measure. It uses Bayes theorem to infer the unknown parameters of a model. Bayesian inference is useful in many applications which involve reasoning and diagnosis such as game theory, medicine, etc. Bayesian inference is much more powerful in handling missing data and for estimating any uncertainty in predictions.

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Fundamentals of Bayes Theorem

Naïve Bayes Model relies on Bayes theorem that works on the principle of three kinds of probabilities called prior probability, likelihood probability, and posterior probability.

Prior Probability

It is the general probability of an uncertain event before an observation is seen or some evidence is collected. It is the initial probability that is believed before any new information is collected.

Likelihood Probability

Likelihood probability is the relative probability of the observation occurring for each class or the sampling density for the evidence given the hypothesis. It is stated as *P* (Evidence | Hypothesis), which denotes the likeliness of the occurrence of the evidence given the parameters.

Posterior Probability

It is the updated or revised probability of an event taking into account the observations from the training data. *P* (Hypothesis | Evidence) is the posterior distribution representing the belief about the hypothesis, given the evidence from the training data. Therefore,

Posterior probability = prior probability + new evidence

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Classification Using Bayes Model

$$P\,(\text{Hypothesis } h \mid \text{Evidence } E) = \frac{P(\text{Evidence } E \mid \text{Hypothesis } h)\ P(\text{Hypothesis } h)}{P(\text{Evidence } E)} \qquad (8.1)$$

Posterior Probability α Prior Probability $\times$ Likelihood Probability

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Maximum A Posteriori (MAP) Hypothesis, hMAP

Given a set of candidate hypotheses, the hypothesis which has the maximum value is considered as the *maximum probable hypothesis* or *most probable hypothesis*. This most probable hypothesis is called the Maximum A Posteriori Hypothesis h_{MAP}. Bayes theorem Eq. (8.1) can be used to find the h_{MAP}.

$$
\begin{aligned}
h_{MAP} &= \max_{h \varepsilon H} P(Hypothesish \mid Evidence\ E) \\
&= \max_{h \varepsilon H} \frac{P(Evidence\ E \mid Hypothesis\ h) P(Hypothesis\ h)}{P(Evidence\ E)} \\
&= \max_{h \varepsilon H} P(Evidence\ E \mid Hypothesis\ h) P(Hypothesis\ h) \qquad (8.2)
\end{aligned}
$$

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Maximum Likelihood (ML) Hypothesis, hML

Given a set of candidate hypotheses, if every hypothesis is equally probable, only $P(E \mid h)$ is used to find the *most probable hypothesis*. The hypothesis that gives the maximum likelihood for $P(E \mid h)$ is called the Maximum Likelihood (ML) Hypothesis, h_{ML}.

$$h_{ML} = \max_{h \in H} P(Evidence\ E \mid Hypothesis\ h) \quad (8.3)$$

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Naïve Bayes Algorithm

Algorithm 8.1: Naïve Bayes

1. Compute the prior probability for the target class.
2. Compute Frequency matrix and likelihood Probability for each of the feature.
3. Use Bayes theorem Eq. (8.1) to calculate the probability of all hypotheses.
4. Use Maximum A Posteriori (MAP) Hypothesis, h_{MAP} Eq. (8.2) to classify the test object to the hypothesis with the highest probability.

ZERO PROBABILITY ERROR

Zero-probability error can be solved by applying a smoothing technique called Laplace correction which means given 1000 data instances in the training dataset, if there are zero instances for a particular value of a feature we can add 1 instance for each attribute value pair of that feature which will not make much difference for 1000 data instances and the overall probability does not become zero.

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Brute Force Bayes Algorithm

Applying Bayes theorem, Brute Force Bayes algorithm relies on the idea of concept learning wherein given a hypothesis space H for the training dataset T, the algorithm computes the posterior probabilities for all the hypothesis $h_i \in H$. Then, Maximum A Posteriori (MAP) Hypothesis, $\boldsymbol{h}_{MAP}$, is used to output the hypothesis with maximum posterior probability. The algorithm is quite expensive since it requires computations for all the hypotheses. Although computing posterior probabilities is inefficient, this idea is applied in various other algorithms which is also quite interesting.

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Bayes Optimal Classifier

Bayes optimal classifier is a probabilistic model, which in fact, uses the Bayes theorem to find the most probable classification for a new instance given the training data by combining the predictions of all posterior hypotheses, This is different from Maximum A Posteriori (MAP) Hypothesis, h_{MAP} which chooses the maximum probable hypothesis or the most probable hypothesis.

Here, a new instance can be classified to a possible classification value C_i by the following Eq. (8.4).

$$= \max_{C_i} \sum_{h_i \in H} P(C_i \mid h_i)\, P(h_i \mid T) \qquad (8.4)$$

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Gibbs Algorithm

The main drawback of Bayes optimal classifier is that it computes the posterior probability for all hypotheses in the hypothesis space and then combines the predictions to classify a new instance.

Gibbs algorithm is a sampling technique which randomly selects a hypothesis from the hypothesis space according to the posterior probability distribution and classifies a new instance. It is found that the prediction error occurs twice with the Gibbs algorithm when compared to Bayes Optimal classifier.

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Naïve Bayes Algorithm For Continuous Attributes

There are two ways to predict with Naive Bayes algorithm for continuous attributes:

1. Discretize continuous feature to discrete feature.
2. Apply Normal or Gaussian distribution for continuous feature.

GAUSSIAN NAIVE BAYES ALGORITHM

IN GAUSSIAN NAIVE BAYES, THE VALUES OF CONTINUOUS FEATURES ARE ASSUMED TO BE SAMPLED FROM A GAUSSIAN DISTRIBUTION.

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Other Popular Types of Naive Bayes Classifiers

Bernoulli Naive Bayes Classifier

Bernoulli Naive Bayes works with discrete features. In this algorithm, the features used for making predictions are Boolean variables that take only two values either 'yes' or 'no'. This is particularly useful for text classification where all features are binary with each feature containing two values whether the word occurs or not.

Multinomial Naive Bayes Classifier

This algorithm is a generalization of the Bernoulli Naive Bayes model that works for categorical data or particularly integer features. This classifier is useful for text classification where each feature will have an integer value that represents the frequency of occurrence of words.

Multi-class Naïve Bayes Classifier

This algorithm is useful for classification problems with more than two classes where the target feature contains multiple classes and test instance has to be predicted with the class it belongs to.

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Summary

1. Probabilistic learning uses the concept of probability theory that describes how to model randomness, uncertainty and noise to predict the future events.
2. It is a tool for modelling large datasets and uses Bayes rule to infer unknown quantities, predict and learn from data.
3. Bayesian learning differs from probabilistic learning in a way that it uses subjective probabilities to infer parameters of a model.
4. Probabilities are used to denote the degree of belief in the occurrence of an event.
5. Bayes' theorem uses conditional probability that can be defined via Joint probabilities where $P(A \mid B) = P(A, B)/P(B)$.
6. Bayes theorem is used to select the most probable hypothesis from data, considering both prior knowledge and posterior distributions.
7. Bayes theorem helps in calculating the posterior probability for several hypotheses and selects the hypothesis with the highest probability.
8. Naïve Bayes Algorithm is a supervised binary class or multi-class classification algorithm that works on the principle of Bayes theorem.
9. Zero probability error with Naïve Bayes Model can be solved by applying a smoothing technique called Laplace correction.
10. Naïve Bayes Algorithm for Continuous Attributes can be solved using Gaussian distribution.
11. Other popular types of Naive Bayes Classifier are Bernoulli Naive Bayes Classifier, Multinomial Naive Bayes Classifier and Multi-class Naïve Bayes Classifier, etc.

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